

Project 8

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MATH-4309-110 Advanced Analysis
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Decemebr 11, 2025

1 Course Certificate

Include a screenshot of the completed course: *Machine Learning Foundations: Calculus*. [Sem23]

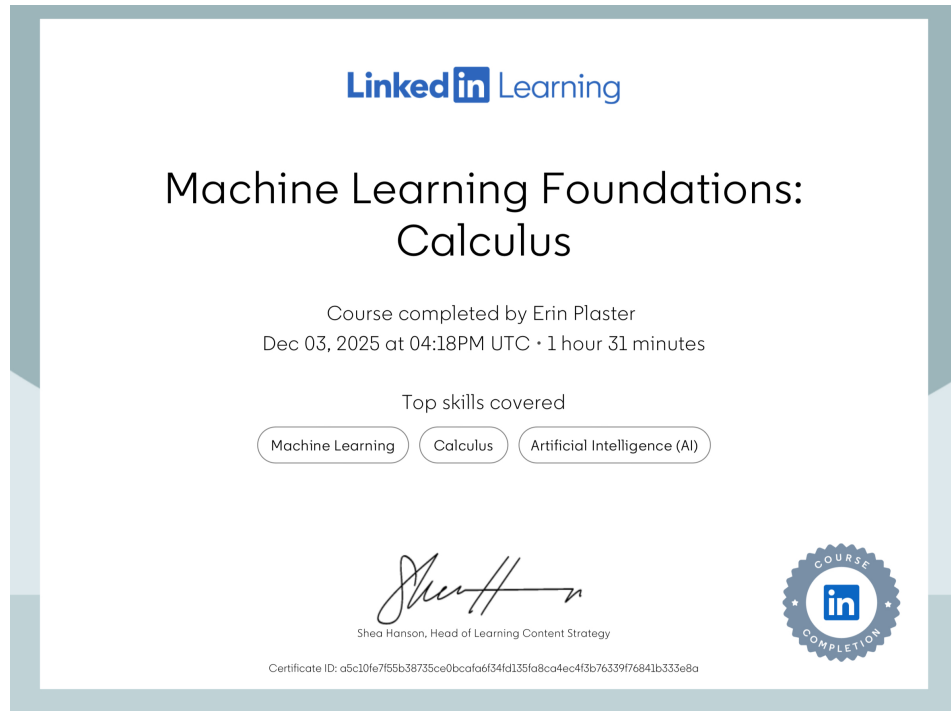


Figure 1: Course certificate for Machine Learning Foundations: Calculus

2 Technical Reflection

The two sections submitted for the capstone project illustrate my understanding of mathematical concepts that relate to analytical calculus and data visualization. Project A, concentrating on Riemann sums and Riemann integration, is an analytical discussion that is important to calculus and my mathematics education in the future. Project B concentrates on both an image classifier application and a linear regression model. I decided to leave out the Fundamental Theorem of Calculus from Project B because the project was already lengthy and it did not fit into the data visualization structure that was requested.

My original projects both needed refinement and organization in order to be included in the capstone. I followed the structure provided by the rubric and expanded sections or included them if they were not part of my original submission, such as conclusions, deeper analysis, research questions, exploratory data, and some overall restructuring to comply with the project goals. Restructuring my project gave me additional understanding related to both Riemann sums and classification. I was able to look more closely at Random Forests and consider where that type of classification might need either revision or additional feature extraction assistance in order to be used in a real-world scenario. I was also introduced to additional mathematical formulations related to Random Forests during my research that I would like to work with in the future.

The project introduced a more formal tone and structure to my mathematical writing and I feel that I have a deeper understanding of what is required for a professional submission. I gained new skills related to programming in R and Python and was introduced to machine learning models and structure. The project has made me consider how I can include machine learning in my education and how it might be reflected in future work. This portfolio supports my future goals because I will have a centralized location for my work that collaborators or employers can review.

References

- [Sem23] Terezija Semenski. *Machine Learning Foundations: Calculus*. Mar. 2023. URL: <https://www.linkedin.com/learning/machine-learning-foundations-calculus/>.